

Cole Kennedy

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EDUCATION

University of South Carolina

Expected Dec. 2027

Bachelor of Science in Computer Science; GPA: 3.93

Columbia, SC

- Academic Scholar - Distinction Award, Capstone Scholar, President's Honor List 3x
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Computer Architecture, Advanced Programming Techniques (C++), Discrete Mathematics, UNIX/Linux Fundamentals

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Present

Evolv Technology

Waltham, MA

- Owned the Collimator Alignment feature end-to-end for Evolv's eXpedite system, rebuilding a legacy laptop tool into an integrated React/TypeScript interface for technician alignment of live X-ray detector hardware.
- Engineered a real-time gRPC-streaming and Electron IPC pipeline ingesting dual-energy X-ray frames, computing per-board gain across 8 boards $\times$ 64 columns against a simulated hardware backend.
- Modeled detector gain (reference / raw-count) and mapped host statistical limits into native count space, replacing estimated thresholds with production values matching the firmware's first/last-board acceptance test.
- Designed custom SVG visualizations and a multi-phase workflow (pre-check $\rightarrow$ aligning $\rightarrow$ pass/fail) rendering live per-column traces, limit bands, and edge-detector exclusion for real-time hardware feedback.
- Integrated the UI with embedded-Linux C++ services via gRPC/Protobuf, orchestrating diagnostics setup mode through database-flag writes, reboot control, and system-status-driven navigation.

Software Engineer Intern

Oct. 2025 – May 2026

Knead

Remote

- Built a Python code generation pipeline to translate user-defined trading strategies into executable backtesting code, enabling flexible composition of event-driven strategies from reusable strategy components
- Designed an authentication architecture using AWS Cognito and implemented JWT-based authorization backed by Supabase, securing backend APIs and enforcing user-level data access across services
- Developed a Python simulation and validation framework for generated trading strategies, covering edge cases through controlled execution and automated checks
- Integrated asynchronous backtesting workflows with backend APIs, reducing end-to-end strategy evaluation time by $\sim 40\%$ through improved data pipeline efficiency and parallel execution
- Refactored live strategy execution services into a stateless, multi-tenant backend with persistent executor state, per-user isolation, and recovery of active strategies after server restarts

Instructional Assistant

Aug. 2025 – Present

University of South Carolina

Columbia, SC

- Led weekly peer instruction sessions on advanced C++ topics including object-oriented design, pointers, templates, and memory management, with emphasis on correctness, performance, and debugging
- Designed and taught interactive coding assignments that integrate data structures, algorithmic analysis, and debugging strategies to help students apply theory to practical programming problems

PROJECTS

Database Engine | C++, STL, React, TypeScript

2026

- Built a disk-backed key-value database engine in C++ with page-based storage, custom record serialization, overflow page handling, persistent B+ tree indexing, buffer pool caching, and crash-aware file validation
- Developed a REST API and React dashboard for record management, B+ tree visualization, and database metrics.

Asynchronous Job Processing Service | Python, FastAPI, Redis, PostgreSQL, Docker

2025

- Built an async job service using Redis queues and workers to decouple API handling from background execution
- Implemented PostgreSQL job state management with retries, idempotency, and fault-tolerant lifecycle handling

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, HTML, CSS

Frameworks/Libraries: React, Electron, FastAPI, Flask, Tailwind CSS

Technologies: gRPC, Protobuf, PostgreSQL, Supabase, Redis, Docker, Git, Linux, AWS Cognito